Andreas W. Neitzel

PhD candidate in astronomy. Machine learning for Galactic archaeology.

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Education

PhD **Faculdade de Ciências da Universidade do Porto and IA/CAUP**, Astronomy — Porto, Portugal, Sept 2022 – present
- Thesis: Machine Learning Applied to Galactic Archaeology
- Advisors: Tiago L. Campante, Diego Bossini, Andrea Miglio

MSc **Faculdade de Ciências da Universidade do Porto**, Astronomy and Astrophysics — Porto, Portugal, Sept 2020 – Dec 2022
- Thesis: Determination of Age and Mass for Seismic Stars in the Ariel Input Catalog
- Final grade 16/20

BSc **Faculdade de Ciências da Universidade do Porto**, Physics — Porto, Portugal, 2017 to 2020

Research positions

INAF-OAS Bologna, Visiting PhD Student, INAF Osservatorio di Astrofisica e Scienza dello Spazio — Bologna, Italy, Sept 2025 – Mar 2026
Six-month secondment in A. Miglio's group building simulation pipelines for the proposed HAYDN mission.

IA/CAUP, PhD Researcher, Stellar Astrophysics Group — Porto, Portugal, Sept 2022 – present
Doctoral research on machine-learning methods for Galactic archaeology with Gaia, APOGEE and asteroseismic data.

Publications

Dissecting stellar populations with manifold learning. I. Validation of the method on a synthetic Milky Way-like galaxy — Mar 2025
A. W. Neitzel, T. L. Campante, D. Bossini, A. Miglio
10.1051/0004-6361/202451718 (Astronomy and Astrophysics, 695, A243)

Expanding the frontiers of cool-dwarf asteroseismology with ESPRESSO: detection of solar-like oscillations in the K5 dwarf epsilon Indi — Mar 2024
T. L. Campante, H. Kjeldsen, Y. Li, M. N. Lund, *et al. including A. W. Neitzel*
10.1051/0004-6361/202449197 (Astronomy and Astrophysics, 683, L16)

Ariel stellar characterisation. I. Homogeneous stellar parameters of 187 FGK planet host stars: description and validation of the method — July 2022
L. Magrini, C. Danielski, D. Bossini, *et al. including A. W. Neitzel*
10.1051/0004-6361/202243405 (Astronomy and Astrophysics, 663, A161)

Talks and posters

- Oral: JANUS: a neural network for APOGEE-like chemistry of the TESS asteroseismic red giants from Gaia XP spectra, with per-star reliability. IMAS13, 13th Iberian Meeting on Asteroseismology, Cádiar (Granada), Spain, 2026-10.
- Poster: HAYDNsim: Forecasting photometric noise in the crowded fields that defeat current missions. TASC10/KASC17, 10th TESS and 17th Kepler Asteroseismic Science Consortium Workshop, Aarhus, Denmark, 2026-07.
- Oral: Contrastive deep learning for structure-preserving chemistry of TESS asteroseismic red giants from Gaia XP spectra. TASC10/KASC17 workshop, Aarhus, Denmark, 2026-07.
- Oral: Identification of stellar populations through latent space projection. TASC8/KASC15, 8th TESS and 15th Kepler Asteroseismic Science Consortium Workshop, Porto, Portugal, 2024-07.

Grants, fellowships and computing allocations

PhD research fellowship under the FCT project ArqueoGal (CIAAUP-17/2026-BI-D) Sept 2026 – Aug 2027
Fellow. FCT, through project 2024.15303.PEX, hosted at IA. Twelve-month PhD research fellowship funded by the ArqueoGal project.

FCT Exploratory Research Project ArqueoGal: Unraveling the evolutionary history of the Milky Way with Galactic Archaeology (2024.15303.PEX) Feb 2026 – Aug 2027
Participant. Fundação para a Ciência e a Tecnologia. Named participant and developer of the project's machine-learning pipelines, with allocations on the Stellar Population Classification and Galactic Modeling tasks.

FCT Doctoral Fellowship (2025.01843.BD) Sept 2025 – Aug 2026
Fellow. Fundação para a Ciência e a Tecnologia. Competitive national doctoral fellowship, held for twelve months.

Awards

Best Master's Student in Astronomy and Astrophysics Faculdade de Ciências da Universidade do Porto 2023

Software

Starfold v0.2.0, MIT, sole developer Apr 2026 – present
Python package that finds groups in a numeric table without labels, using UMAP, HDBSCAN and Optuna, and reports how far to trust them: recurrence across refits, named reliability flags and an optional Monte Carlo calibration test; extends Neitzel et al. 2025.

Skills

Python engineering for research software: Typed Python packages with tests, continuous integration, locked environments, provenance stamps, Sphinx documentation.

Probabilistic deep learning: conditional normalising flows, ensembles, mixture uncertainties: Designed and trained an ensemble of flow-based networks that returns a joint posterior over 25 stellar labels per star, with the uncertainty of each value and the disagreement between networks.

Uncertainty calibration and reliability: isotonic calibration, gradient boosting, selection tests, coverage: Built the per-star reliability layer of JANUS: a calibrated probability that a prediction is wrong (gradient boosting with isotonic calibration, cross-fitted), Poisson-binomial selection tests, and PIT-based interval coverage checks.

Out-of-distribution detection: nearest-neighbour novelty in a Gaussianised input space: A novelty score per star from the distance to the training set in the rank-transformed input space, validated on survey samples outside the training selection.

Data engineering at scale: archive ingestion, Parquet pipelines, a manifest-driven data store with provenance: Designed the central data store and the ingestion of whole Gaia releases (archive queries and bulk files to columnar shards) with manifest-resolved datasets, anti-duplication rules and provenance on every product.

Unsupervised learning and manifold methods: UMAP, HDBSCAN, Optuna search, stability and recurrence measurement, matched-null calibration.

High-performance computing: Slurm on ARM, x86 and GPU partitions of the Deucalion supercomputer and the UniBo clusters: JANUS training and production scoring, HAYDN and PLATO photometric-noise runs, memory-bounded process fan-out, submission gates and per-job cost ledgers; CUDA and RAPIDS workflows.

Survey data and stellar modelling tools: Gaia DR3 (astrometry, RVS, XP spectra, GSP-Spec), APOGEE, LAMOST, GALAH, 2MASS, TESS and Kepler/K2 asteroseismology; astropy, galpy, TOPCAT, ADQL, PARAM, MESA, TRILEGAL, MWDUST.